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RESEARCH ARTICLE

FarmSTEPS, a model for spatial-temporal exploration of diversified and sustainable crop sequence plans for crop farms

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Abstract

Crop diversification could enhance economic and environmental sustainability of crop farming, but achieving this requires careful planning that accounts for crop interrelationships in a rotation or spatial layout. Several models have been devised to design diversified crop rotations or crop spatial layouts, but they often fail to capture farm settings, production constraints, and sustainability objectives specific to a farm at both temporal and spatial scales. To fill this gap, we developed a novel rule-based whole-farm planning model, FarmSTEPS, for systematically exploring diversified and sustainable farm crop sequence plans in a spatial-temporally explicit manner. FarmSTEPS comprises multiple farm characterization modules, an algorithm for generating farm crop sequence plans, and a multi-attribute decision-aid tool. Users first characterize farm configurations and plot properties, and parameterize candidate crops according to the actual situation of farms. Subsequently the algorithm generates and evaluates all feasible crop sequence plans for each plot from the first planning period until the end of the planning window, subject to user-defined sequence- and farm-level constraints. Eventually, the model suggests a set of superior farm plans using the multi-attribute decision-aid tool. We illustrate the model with a Chinese wheat-maize farm to explore future three-year cropping plans targeting five sustainability objectives: maximizing gross margin and minimizing four environmental impacts (groundwater depletion, pesticide input, aquatic eutrophication, GHG emission). The model generated 213,679 feasible plans, with 128 being Pareto-optimal in terms of the five objectives. Overall, relative to the wheat-maize baseline, these optimal plans increased gross margin by 3-50% while mitigating environmental impacts by 10-56%, largely through the inclusion of tuber and oil crops, though trade-offs existed between economic and environmental benefits. The case study demonstrates that FarmSTEPS is an effective tool for designing diversified and sustainable crop sequence plans tailored to farm-specific conditions while informing stakeholders on potential benefits and trade-offs among sustainability objectives.

Keywords Crop diversification · Farming system redesign · Pareto optimality · Profitability · Environmental sustainability · North China Plain

1 Introduction

Sustainable crop farming is required to produce sufficient crop products while ensuring farm incomes and minimizing environmental externalities (Cui et al. 2018). Many major crop-producing regions around the world have been experiencing crop specialization and intensification in the past few decades, with farmlands dominated by single or a few crop species, particularly major cereals (Ramankutty et al. 2018). This trend contributed to increased agricultural productivity in terms of dietary energy but caused various farming sustainability challenges in both the economic and environmental domains, such as undesired and unstable operating profits, overexploitation of freshwater (particularly groundwater), exposure of humans and the environment to agrochemicals, greenhouse gas (GHG) emissions, and agro-biodiversity loss (Campbell et al. 2017; Chakraborti et al. 2023), threatening economic well-being and environmental health. It is therefore crucial to transform currently intensified and specialized crop farming systems toward increased sustainability.

Crop diversification is deemed an effective strategy to improve the multi-attribute sustainability of crop farming (Hufnagel et al. 2020). From a temporal perspective, it

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presents itself in the form of diversified crop sequences, i.e., cultivating a sequence of different crops on the same piece of land, while from a spatial perspective, crop diversification results in spatial cropping plans with different crops across different pieces of land every year (Dury et al. 2012). Sustainability benefits of crop diversification have been found in many previous studies. A meta-analysis and a systematic review both indicated that diversified cropping systems could deliver higher financial returns relative to simplified systems, often via introducing high-profit crops such as vegetables and root crops (Sánchez et al. 2022; Feliciano 2019). Furthermore, multiple empirical and meta-analysis studies have demonstrated that diversification could reduce the reliance on agricultural resource inputs, including groundwater, fertilizer, pesticide, and fossil energy, and mitigate associated environmental impacts by incorporating resource-efficient crops such as beans, coarse cereals, and some oil crops (Boeraeve et al. 2020; Hunt et al. 2020; Guinet et al. 2023; Yang et al. 2024). The successful cases in these studies highlight that achieving the sustainability benefits of diversified systems requires careful, reasoned planning.

Models serve as an efficient tool for planning sustainable and diversified crop production systems because they can handle, systematically and transparently, many factors for cropping system innovation, including various candidate crops, system constraints, and sustainability objectives (Dury et al. 2012). Several models have been devised to facilitate the planning of diversified systems either temporally or spatially. Among them, crop rotation generator models such as ROTAT+ and ROTOR can create all agronomically viable crop rotations at the plot scale and evaluate their multi-attribute sustainability, given a list of candidate crops and rotation rules (Dogliotti et al. 2003, 2004; Bachinger and Zander 2007). However, the generated sustainable rotations are often difficult to implement on real farms since the models do not account for the farm-specific situations and demands, for example, the heterogeneity in size and properties of plots within a farm, resource accessibility and availability at a farm, and the whole-farm performance objectives pursued by farmers and other stakeholders. Moreover, these models were built merely for fixed rotation systems, which follow a cyclical pattern of component crops, rather than generating flexible and non-cyclical crop sequences for the future (Castellazzi et al. 2008). This model limitation might constrain the scope of sustainability improvement delivered by temporal crop diversification for farms (Dury et al. 2013).

Some whole-farm planning models are able to allocate crops or crop sequences to different farm plots to improve farm-level sustainability objectives using optimization techniques such as linear programming and heuristic algorithms (Dogliotti et al. 2005; Groot et al. 2012; Sempore et al. 2015; Capitanescu et al. 2017). But there are several limitations to these models: 1) primarily optimizing a single objective (usually profits) rather than multiple ones when using linear programming; 2) overlooking key agronomic constraints such as disallowed crop successions or maximum planting frequency of crops in a crop sequence; 3) missing period-by-period assessment and constraints on farm cropping plans, e.g., the minimum farm profit in each year, making the resulting diversified farm plans less practical for farmers (Kaim et al. 2018); and 4) lacking capabilities to create non-cyclical and flexible crop sequences over planning horizons (Janssen and Van Ittersum 2007). There is still a need for a modelling framework for the spatial-temporal exploration of whole-farm cropping plans based on farm-specific situations and multiple sustainability objectives. This topic remains largely unexplored.

The North China Plain is a major crop-producing region in China and a typical intensified and specialized crop farming region , where over 60% of the farmland is currently devoted to annual double cropping of winter wheat and summer maize (Li and Lei 2021). This cereal-dominated system is highly water- and nutrient-demanding, with an estimated blue water demand of 300 mm/year and nitrogen uptake of 400 kg N/ha/year (Yang et al. 2021; Zhang 2021). To achieve higher productivity, farmers often use a considerable amount of groundwater and fertilizers (Liang et al. 2022). This practice leads to a sharp groundwater table decline and a high N surplus (approximately 250 kg N/ha/year; Zhang 2021), which contributes to waterbody eutrophication (Kong et al. 2016; Zhang et al. 2017). Additionally, Chinese agri-environmental policies target the reduction of chemical pesticides and the mitigation of carbon emissions (MARA 2021). To reduce the environmental burden arising from crop production, the local government encourages farmers to partially replace staple cereals with low-input crops, especially on large-scale farms that have sufficient resources and skills to manage new crops (Xu et al. 2024; Fig. 1). However, farm managers are more concerned about farm gross margin and lack methods for designing profitable yet environmentally benign farm cropping plans tailored to specific situations of their farms. With this need, the typical wheat-maize farm in the North China Plain could serve as an ideal case for model-based exploration of sustainable whole-farm crop sequence plans.

The primary aim of this study is to present a novel model, FarmSTEPS, for exploration of diversified and sustainable crop sequence plans for crop farms aligned with farm-specific conditions and requirements. FarmSTEPS is a whole-farm modelling framework capable of systematically generating desired farm cropping plans in a spatial-temporally explicit manner by integrating farm characterization modules, an exhaustive whole-farm crop sequence planning algorithm, and a multi-attribute decision-aid tool. The second aim is to illustrate the application of this model by exploring alternative options for a wheat-maize family

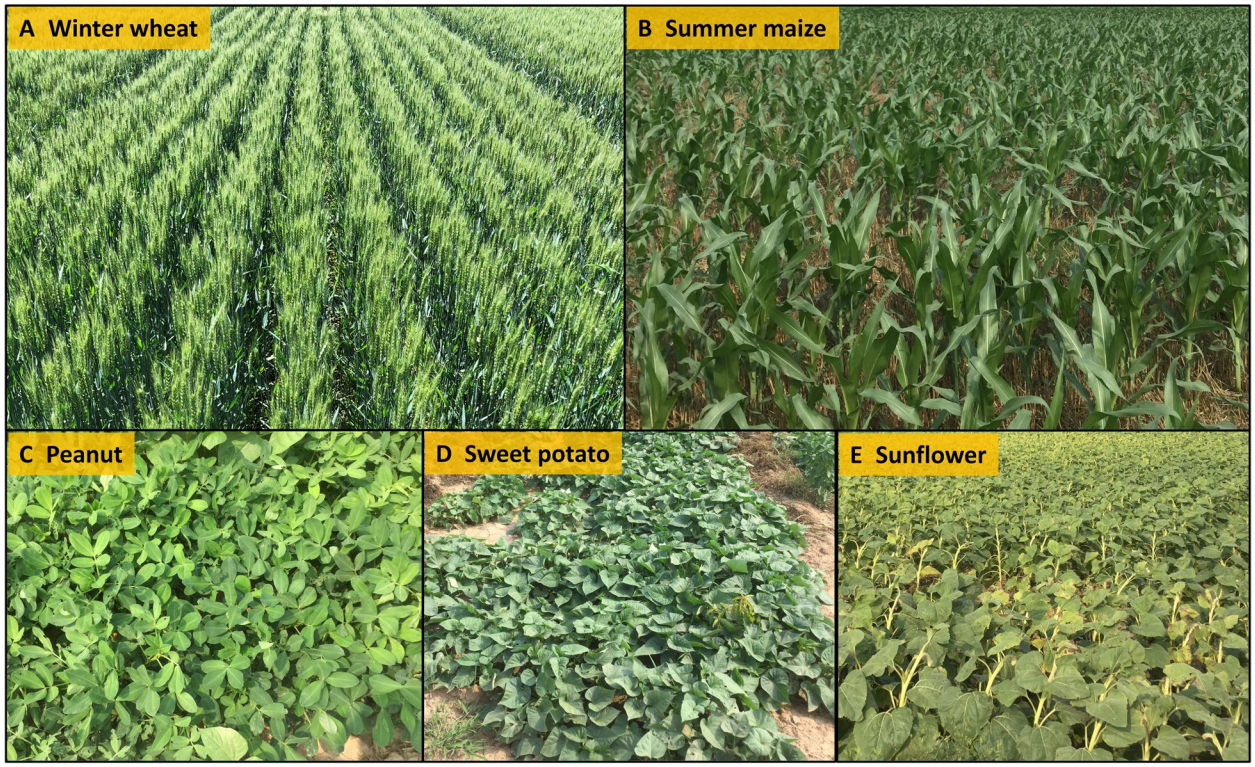


Fig. 1 Photographs of widely cultivated crops (winter wheat and summer maize; **A**, **B**) and less commonly cultivated crops (peanut, sweet potato and sunflower; **C**–**E**) in the North China Plain. The annual winter wheat-summer maize annual double cropping system dominates regional farmland, while those low-input minor crops, including oil and tuber/root crops, offer opportunities to diversify local arable farming systems for enhancing both economic and environmental sustainability (photo credit: Zhengyuan Liang and Zhan Xu).

farm in the North China Plain when developing future three-year cropping plans with five sustainability objectives, i.e., maximizing gross margin and minimizing four environmental impacts. This application allows us to assess the potential benefits and trade-offs among sustainability objectives when transitioning to new farm plans. The model flexibility allows it to be applied to other farms, adapting to specific farm settings, production constraints, and sustainability objectives.

2 Model

We developed a rule-based model, FarmSTEPS (Spatio-Temporal Exploration ProcedureS for crop allocation on a Farm, Fig. 2) to systematically generate and evaluate feasible crop sequence plans for plots within a given crop farm in a spatial-temporally explicit manner, subject to both sequence- and farm-associated constraints. This was done by integrating multiple modules for farm characterization and stakeholder expectations (section 2.1) with a whole-farm crop sequence planning algorithm (section 2.2). Model users could apply FarmSTEPS by parameterizing these modules in terms of the real situation and objectives and constraints of the focus farm and accordingly obtain a set of possible cropping plans and their performance as model output. Then a multi-attribute decision-aid tool, Pareto ranking, efficiently identifies the superior solutions from the entire set of possible plans in terms of sustainability objectives of interests, which could inform the stakeholders' decision-making process (Section 2.3). We programmed the model software in the C# language in Microsoft Visual Studio.

2.1 FarmSTEPS modules

2.1.1 Farm configuration module

Since each plot within a farm is the basic unit for crop sequence planning, the initial step in applying the model is configuring the farm by loading the farm plot map in an ESRI shape file, where each plot is represented by a GIS polygon. The key aim of this module is to read, store, and display the farm plot configuration along with relevant geographical attributes, e.g., the position, size, and slope per plot.

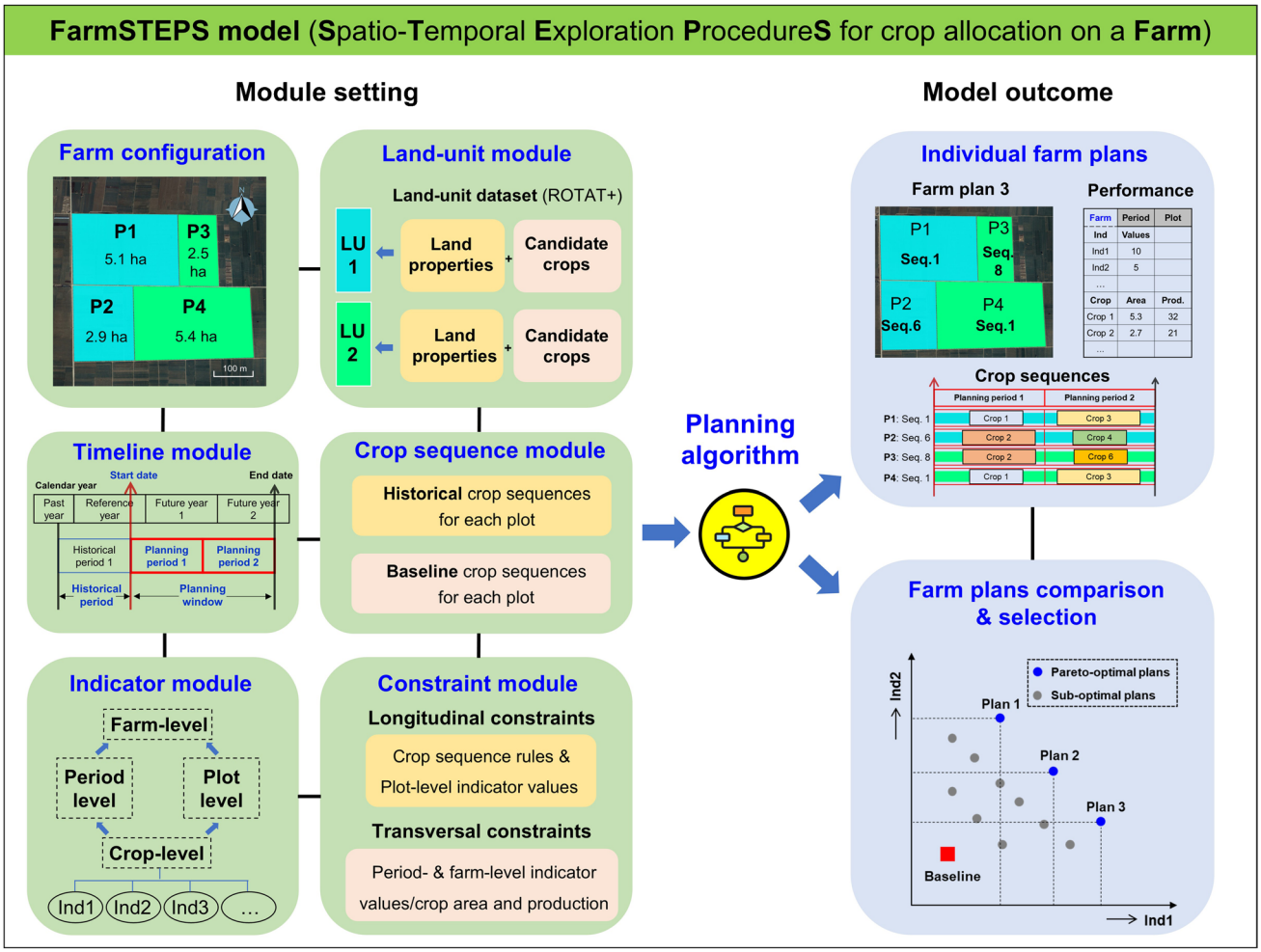


Fig. 2 The modeling framework of the FarmSTEPS model. Model users firstly characterize the farm in terms of plot configuration, land-units, and crop sequences (see Sections 2.1.1, 2.1.2, 2.1.4 in the main text), and define the timeline, indicators, and constraints (see Section 2.1.3, 2.1.5, 2.1.6 in the main text) for the generation and assessment of alternative farm cropping plans by setting six interconnected modules (farm configuration module, land-unit module, timeline module, crop sequence module, indicator module and constraint module; Sections 2.1 in the main text). Next, the planning algorithm creates all feasible farm crop sequence plans in a spatial-temporally explicit way for the whole farm over the whole planning window (Section 2.2 in the main text). Users eventually can check the indicator performance and crop sequences of individual farm cropping plans, compare the performance across the entire set of feasible cropping plans, and identify the better-performing plans (Section 2.3 in the main text). This framework was presented with a hypothetical farm where P1-P4 denote four plots, LU1-2 denote two land-units, Ind1-Ind3 denote the indicators of interest and Seq. 1, Seq. 6, and Seq. 8 denote three crop sequences used in a created farm plan (Plan 3). Prod. denotes the production of corresponding crops.

2.1.2 Land-unit module

Each plot should be linked with a specific land-unit, which describes characteristics of the plot such as soil properties and includes a list of candidate crops that can be grown on the plot. This allows to consider the differences in environmental conditions, crop selection, and crop performance among plots in a farm. For example, some of the plots might not have access to irrigation and therefore some drought-sensitive crops are banned and the candidate crop yields in these plots would be lower than those in irrigated plots. The land-units are set in the ROTAT+ software that is connected to FarmSTEPS. ROTAT+ is normally used as a tool for field-level crop rotation generation (Liang et al. 2023) while acting as a module in FarmSTEPS to record the land and crop information. Here, users need to parameterize each candidate crop with information that is essential for generating whole-farm crop sequence plans and calculating their indicator performance (see Section 2.1.5), including planting and harvesting dates, agricultural resource inputs (e.g., labor requirement, fertilizer application, pesticide input and irrigation water consumption), and crop and pollutant output (e.g., yield, revenue, nutrient losses, GHG emissions). Since assessment indicators may vary across case farms,

the model is designed to flexibly accommodate user-defined input-output attributes for each crop.

2.1.3 Timeline module

To reflect the actual farmers' dynamic decision-making processes, several timeline parameters should be set for farm planning. The first one is the starting date of planning, determined based on when farmers would change the current crop arrangement or the start of the cropping season. Secondly, the length of the planning period, reflecting how often farmers assess farm performance and make cropping decisions, must be defined. One or multiple crops can be sown and harvested per planning period. This period typically spans one year but can vary, ranging from shorter (e.g., one quarter) to longer (e.g., multiple years). The third parameter is the number of planning periods for crop sequence generation, also known as the "planning window" in FarmSTEPS. Additionally, users can specify the length of historical crop sequences prior to the starting date of farm planning, enabling the model to account for the cropping history of each farm plot. All of these timeline parameters are applied uniformly across all plots within a farm.

2.1.4 Crop sequence module

With defined timeline parameters, this module allows to set the historical and baseline crop sequences for each plot in a farm. Setting historical crop sequences incorporates the influence of past crops on future crop planning in the model. Baseline crop sequences represent the farmers' original idea for arranging crop sequences across plots in the planning window, providing a reference for comparing with novel plans. Users can choose a few candidate crops from the land-units to constitute historical and baseline crop sequences for plots.

2.1.5 Indicator module

Indicator assessment in FarmSTEPS relies on crop input-output relationships so model users can define any indicators of their interest that are calculated in terms of crop input-output dataset to assess farm cropping plans. For instance, dietary nutrient yield from the human-nutrition domain, gross margin and labor use from the socio-economic domain, and groundwater use and GHG emission from the environmental domain. These indicators can be set as objectives for improving sustainability benefits or as constraints for eliminating undesired solutions during planning.

For a given whole-farm crop sequence plan, the module begins with evaluating the crop-level performance ($I_{i,p,t,c}$), i.e., the ith indicator performance of the component crop c of the sequence on the plot p in the tth planning period. If the growing season of a crop spans two planning periods, users can decide whether the indicator values are allocated between periods proportionate to the growing duration or credited to the period when the crop is harvested. The former is suitable for indicators associated with agricultural inputs such as labor input and environmental costs while the latter is suitable for product and economic output-related indicators such as gross margin and food energy production.

Assessment outcomes are subsequently computed at the plot, period and farm level and presented to the user. The indicator performance of each plot ($I_{i,p}$) is calculated as:

$$I_{i,p} = \frac{\sum_t \sum_c I_{i,p,t,c}}{PW} \quad (1)$$

Where PW is the duration of the planning window (in years). The farm performance of indicator i per planning period ($I_{i,t}$) is computed as:

$$I_{i,t} = \frac{\sum_p \sum_c (I_{i,p,t,c} \times A_p)}{\sum_p A_p} \quad (2)$$

Where A_p represents the land area (ha) of plot p. The overall performance of indicator i across the entire planning window ($I_{i,\text{farm}}$) is:

$$I_{i,\text{farm}} = \frac{\sum_p \sum_t \sum_c (I_{i,p,t,c} \times A_p)}{\sum_p A_p \times PW} \quad (3)$$

2.1.6 Constraint module

Various constraints can be formulated in FarmSTEPS to ensure that the generated farm cropping plans comply with the agronomic rules and satisfy the farm-specific demands. Here, all constraints are classified into two types: longitudinal and transversal constraints. The longitudinal constraints are applied to crop sequences on each plot while transversal ones are applied to the whole farm.

Longitudinal constraints Longitudinal constraints, applied at the temporal scale, include sequence-level rules and plot-level constraints.

Sequence-level rules can help to avoid the negative agronomic effects of unreasonable crop successions, for example, yield loss and build-up of soil pathogens. Users can set the following crop sequence rules based on local context: the maximum planting frequency of a certain crop (group), the minimum interval between repeated planting of a certain crop (group), and the disallowed crop succession matrix (Dogliotti et al. 2003). The planned crop

sequences in each plot account for these rules with historical sequences.

Plot-level constraints are used to limit the ith indicator performance of the crop sequence on the plot p within a certain range:

$$I_{i,p,\min} \leq I_{i,p} \leq I_{i,p,\max} \quad (4)$$

Where $I_{i,p,\max}$ and $I_{i,p,\min}$ are the maximum and minimum value of the ith indicator that users enter into the model to introduce their expectations on plot p.

Transversal constraints Transversal constraints, applied at the spatial scale, include period- and farm-level constraints.

Period-level constraints are set to remove the created farm cropping plans that, within a planning period, are unacceptable from the perspective of the indicator performance and/or the harvested area or production of certain crops:

$$I_{i,t,\min} \leq I_{i,t} \leq I_{i,t,\max} \quad (5)$$

$$A_{c,t,\min} \leq A_{c,t} \leq A_{c,t,\max} \quad (6)$$

$$P_{c,t,\min} \leq P_{c,t} \leq P_{c,t,\max} \quad (7)$$

Where $I_{i,t,\max}$ and $I_{i,t,\min}$ are the user-defined maximum and minimum value of the ith indicator at the whole farm in the tth planning period. $A_{c,t,\max}$ and $A_{c,t,\min}$ mean the maximum and minimum harvested area (ha) for crop species c while $P_{c,t,\max}$ and $P_{c,t,\min}$ mean the maximum and minimum production (Mg) for crop species c in the farm for the tth planning period.

Likewise, farm-level constraints can rule out unacceptable farm plans over the entire planning window:

$$I_{i,\text{farm},\min} \leq I_{i,\text{farm}} \leq I_{i,\text{farm},\max} \quad (8)$$

$$A_{c,\text{farm},\min} \leq A_{c,\text{farm}} \leq A_{c,\text{farm},\max} \quad (9)$$

$$P_{c,\text{farm},\min} \leq P_{c,\text{farm}} \leq P_{c,\text{farm},\max} \quad (10)$$

Where for the entire planning window, $I_{i,\text{farm},\max}$ and $I_{i,\text{farm},\min}$ represent the maximum and minimum value of the ith indicator at the whole farm scale, and $A_{c,\text{farm},\max}$, $A_{c,\text{farm},\min}$, $P_{c,\text{farm},\max}$, and $P_{c,\text{farm},\min}$ mean the maximum and minimum harvested area (ha) and production (kg) for crop species c across the plots in the farm, respectively.

2.2 FarmSTEPS planning algorithm

The planning algorithm in the FarmSTEPS software creates a list of feasible crop sequence plans for the whole farm, i.e. 'farm-plans' (across one or more 'planning-periods', which are user defined periods for instance for one year. Generally speaking, the planning procedure initially looks at the first planning-period and generates all possible combinations of consecutive crops ('crop-sequence') for each plot across the whole farm, considering predefined longitudinal and transversal constraints. Then it proceeds to the next planning-period based on the list of possible crop combinations from the previous period and repeats this process until the end of the planning window is reached (Fig. 3), reflecting the farmers' dynamic decision-making processes. The model ultimately provides all feasible farm-plans with their corresponding indicator performance (Fig. 3).

Specifically speaking, the planning commences with the generation of feasible crop sequences per plot in the first planning period. For each plot in the farm, the algorithm constructs a crop-sequence containing a list of crops that have been grown before the start of the first planning-period, i.e. the plot history. Then the last day of the historical crop-sequence is checked to determine whether there is growing time to accommodate a one or more crops in the first planning-period. If growing time is available, the crop-sequence adds the first crop or successive crops from the candidate crop list of the plot. Then the crop sequence rules are applied to the crop-sequence to examine the agronomic feasibility. If this crop-sequence conforms to agronomic rules, it is stored in the collection of crop-sequences for the plot. After that, the added crop(s) in the crop-sequence is(are) emptied to add and examine the next single crop and successive crops from the list. This process runs throughout all candidate crops for the plot and then generates a collection of all agronomically feasible crop-sequences in each plot in the first planning-period.

Subsequently, the algorithm produces all feasible farm cropping plans during the first planning-period. In doing so, a farm-plan by combining feasible crop-sequences across all plots in the farm within the first planning-period. If this farm-plan does not violate the period-level constraints, it is added to the collection of farm-plans for the first planning-period. Similarly, the farm-plan deletes the first combination, and the above process is then repeated to the rest of all combinations of crop-sequences to generate the whole collection of feasible farm-plans. Moreover, a crop-sequence would be marked if used in a feasible farm plan. The crop-sequences that are not marked at the end of the farm plan generation are eliminated from the collection of feasible crop-sequences for a plot.

The planning procedure for creating feasible crop-sequences and farm-plans is also applied for the each of the following planning-periods until the end of the planning window. Here, a crop-sequence for a plot builds on not only its historical sequence but also on one feasible sequence from the previous planning-periods. All crop-sequences for

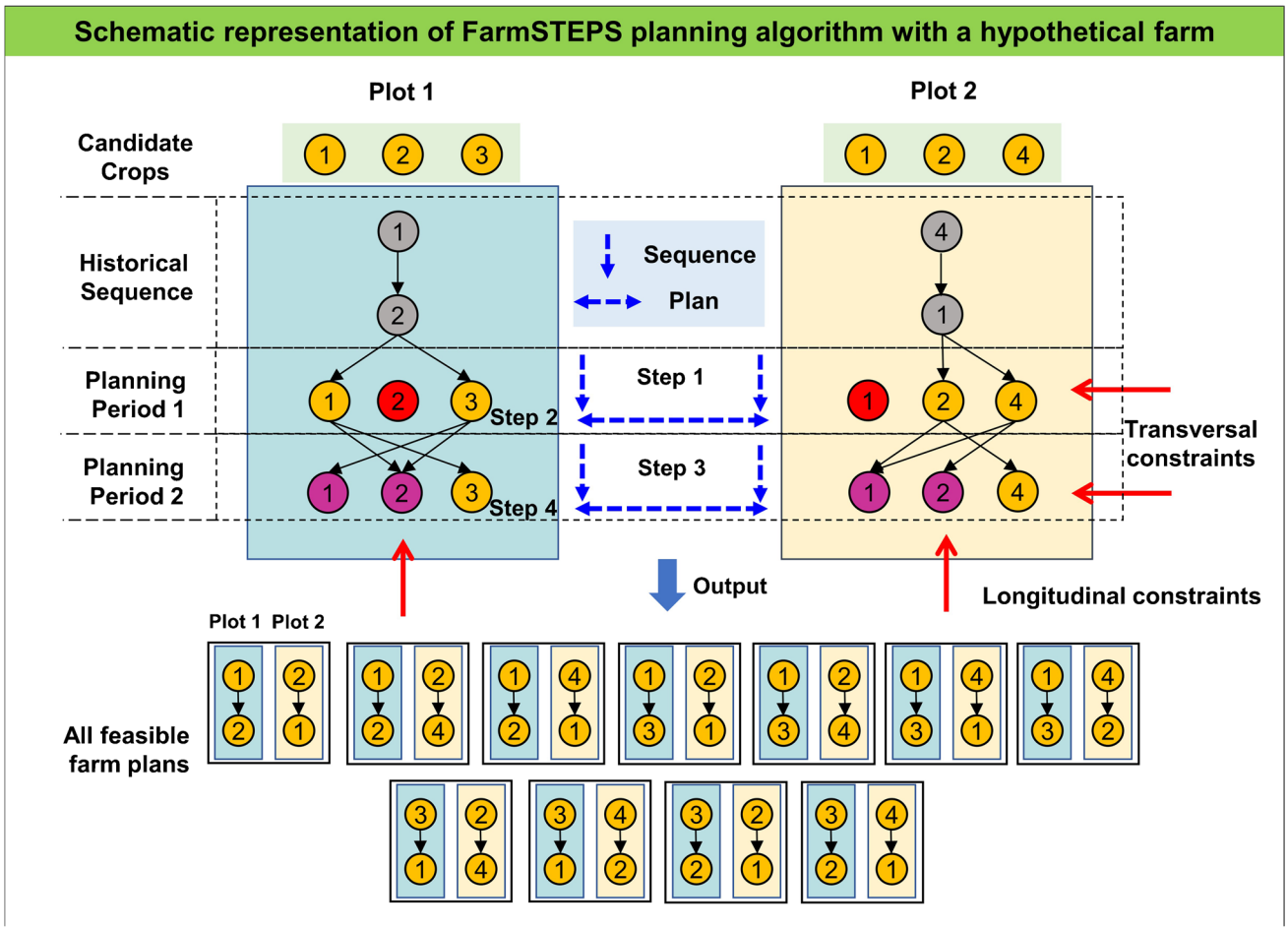
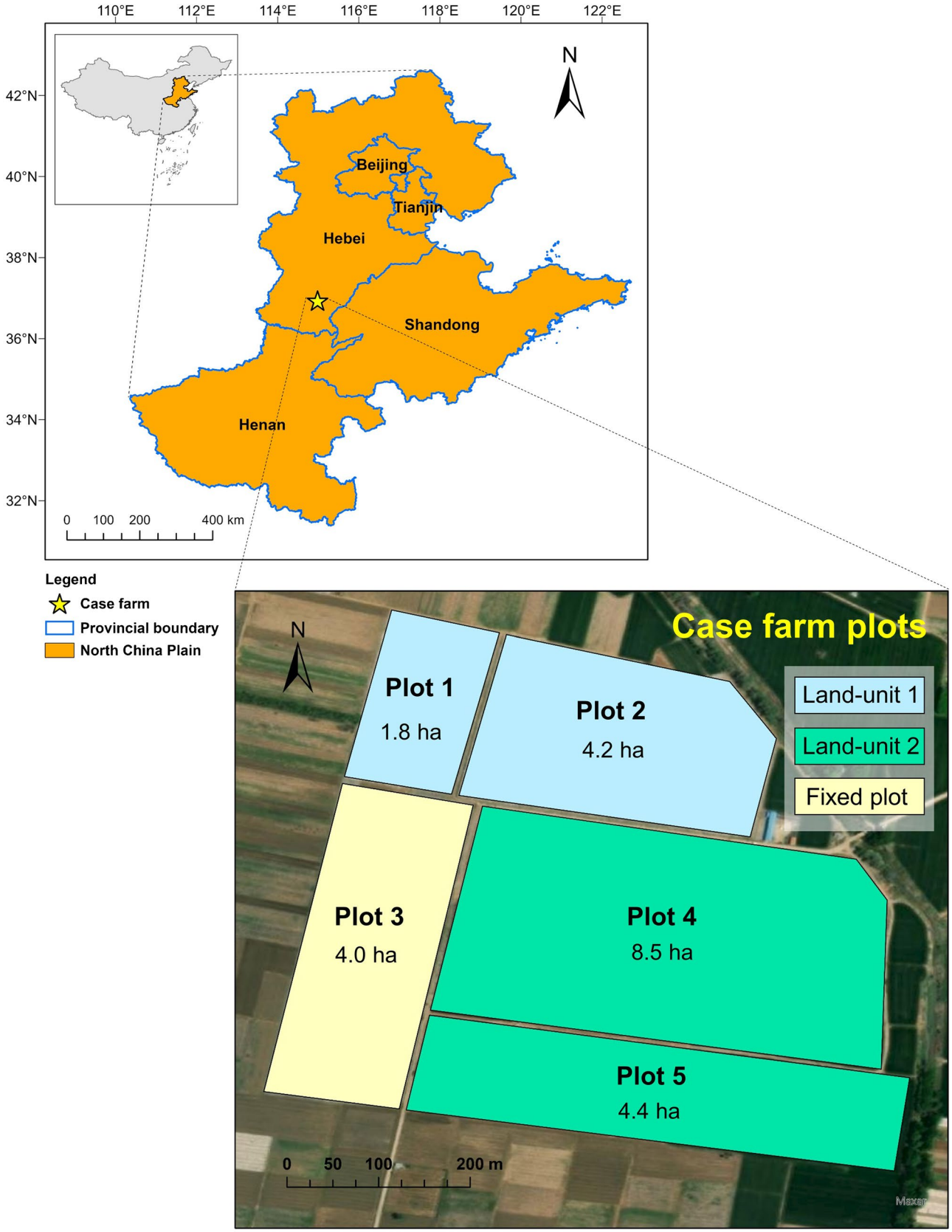


Fig. 3 Sketch of the algorithm for systematic generation of feasible farm-plans. The algorithm is illustrated for generating all feasible plans for a hypothetical farm over two future planning-periods. This farm contains two plots with different candidate crops and historical crop sequences. The vertical blue dashed arrow indicates the generation of crop sequences at the plot level, while the horizontal blue dashed arrow represents the aggregation of sequences into farm-plans. In each planning-period, the model first creates all feasible crop-sequences for each plot, subject to longitudinal constraints (expressed as vertical red arrow; Section 2.1.6 in the main text). For the hypothetical farm, the longitudinal constraint is that the same crop may not be planted consecutively in the same plot. Candidate crops in red color mean their associated crop-sequences were all infeasible due to this longitudinal constraint. Then the model combines feasible crop-sequences between the two plots to produce farm-plans and identifies feasible farm-plans by checking transversal constraints (expressed as horizontal red arrows; Section 2.1.6 in the main text). We set the transversal constraint that each candidate crop is allocated to at most one plot in any given unit planning-period, ensuring that the harvested area of each crop does not exceed the area of a single plot. The above two steps are repeated until the end of the planning window, resulting in a set of all feasible farm-plans (n=11). Candidate crops in rose color mean some plans including them are ruled out due to this transversal constraint.

the plot are systematically added with subsequent crops in the focus period and all agronomically feasible crop sequences are selected and stored, as the algorithm does in the first period. Likewise, all feasible farm-plans for the following period are accordingly obtained with the help of period-level constraints. Furthermore, at the end of the final planning-period, plot- and farm-level constraints remove the farm farm-plans that are unacceptable due to undesirable performance for certain plots and the whole farm across the entire planning window. The remaining farm-plans are deemed feasible and exported as the outcome of the FarmSTEPS planning algorithm.

2.3 Identification of Pareto-optimal farm plans

The FarmSTEPS model typically generates a large set of alternative farm-plans, some of which may underperform in terms of sustainability objectives. These suboptimal farm-plans could therefore be excluded from those recommended to end users. Here, we applied the Pareto ranking method, a non-weighing multi-criteria decision-aid tool, to identify the Pareto-optimal plans from the entire set of generated farm-plans. A plan was ranked as Pareto-optimal when its performance of any one objective cannot be outstripped by another farm-plan without these other plans being inferior to


110°E
112°E
114°E
116°E
118°E
120°E
122°E
42°N
40°N
38°N
36°N
34°N
32°N
N
Beijing
Tianjin
Hebei
Shandong
Henan
0 100 200 400 km
Legend
Case farm
Provincial boundary
North China Plain
Case farm plots
Land-unit 1
Land-unit 2
Fixed plot
Plot 1
1.8 ha
Plot 2
4.2 ha
Plot 3
4.0 ha
Plot 4
8.5 ha
Plot 5
4.4 ha
0 50 100 200 m

◂**Fig. 4** Location and plot configuration of the case farm in the North China Plain for the FarmSTEPS application. The blue and green color of the plots represent two land-units defined in the model, which differed in the set of candidate crops. The plot in yellow was a fixed plot, which was set to be devoted to wheat-maize double cropping over the planning window.

at least one of the other objectives (Groot et al. 2012). This was done in *R* software via the *is_nondominated* function from the "*eaf*" package. While the Pareto ranking has not been directly embedded within the FarmSTEPS software, it serves as an essential component of the overall methodological framework, enabling trade-off analysis and guiding the selection of desirable plans for end users.

3 Case study

3.1 Case farm description

The case farm modeled in FarmSTEPS is a single farm located in Quzhou County, Hebei Province, China (114°98′E, 36°94′N, Fig. 4). This farm is a large-scale family farm with a cropland area of 23 ha. Quzhou has a temperate continental monsoon climate, with a mean annual temperature of 13.1 ℃ and a mean annual precipitation of 430 mm. The farm has a loamy textured calcareous alluvial soil, with a soil bulk density of 1.43 g/cm^3, a pH of 8.2, and a soil organic matter content of 1.57%. Farm managers have been consecutively cultivating only wheat-maize double cropping on the whole farm for at least five years. Double cropping of wheat and maize is a common cropping pattern in the North China Plain due to its high output of cereal grain for human consumption (wheat) and farm animals (maize) and the availability of mechanization.

We selected this farm as a case study to illustrate the capabilities of FarmSTEPS for both practical and methodological reasons.

From a practical perspective, the farmers are currently unsatisfied with the farming profitability and therefore seek to improve farm gross margin by diversifying cropping plans. Furthermore, the local government endeavors to reduce resource and environmental pressures arising from agricultural production such as groundwater overdraft, nutrient pollution, pesticide overuse, and GHG emissions by collaborating with large-scale farms that can set an example for nearby, often smaller, farms. The case farm also encounters practical constraints related to labor availability, market outlets, and on-farm management in the face of altering cropping plans. Thus, the farm acts as a suitable case to use FarmSTEPS to explore diversified farm crop sequence plans towards achieving multi-attribute farming sustainability.

From a methodological perspective, this farm represents multiple dimensions of complexity relevant to whole-farm crop planning, specifically:

Variable plot size: Although the farm consists of only five plots, their sizes vary considerably. This heterogeneity in plot area markedly influences crop allocation decisions and farm-level outcomes;

Diverse candidate crops: The crop portfolio includes a wide range of crops, from staple cereals and tuber crops to oil crops and vegetables, with distinct crop calendars, agronomic requirements, and economic and environmental profiles;

1) Land-unit-specific crop set: The farm includes two land units, each with a different set of candidate crops, reflecting spatial heterogeneity in farm management.
2) Flexible cropping intensity: The one-year unit planning period allows for both single and double cropping systems, increasing the number of feasible crop sequence combinations.
3) Multi-objective assessment: Five sustainability indicators (economic returns, groundwater depletion, pesticide input, eutrophication potential, and GHG emissions) were used to evaluate the cropping plans, capturing trade-offs between farm profitability and environmental impacts.
4) Complex constraints: In addition to crop sequence rules, we considered transversal constraints at the period and farm levels. These include limits on total production and harvested area of crops, as well as thresholds for dietary energy yield, and socioeconomic and environmental indicators.

This case study does not address certain dimensions of complexity, such as spatial variability in soil across plots (Aggarwal et al. 2022) or a longer planning horizon (> three years), which could be tested in the future studies. However, given the multiple forms of complexity represented above, this case provides a robust and sufficiently representative testbed for illustrating the core capabilities of FarmSTEPS.

3.2 Model application for the case farm

3.2.1 Farm, timeline, and crop sequence characterization

The cropland on the case farm consisted of five plots (Plot 1–5; Fig. 4), divided by multiple farm lanes for agricultural machinery transport.

We established two land-units in ROTAT+: one was linked to Plots 1 and 2 while the other one was to Plots 3–5. There was no discrepancy in soil properties and resource availability among the five plots so we did not consider these characteristics in land-units. The candidate crop list for each land-unit was determined through discussion with the farm

managers regarding which crops are suitable for planting on the farm and are acceptable for managers (Table 1). The only difference between the two land-units was the candidate crops because managers were willing to cultivate labor-intensive crops (peanut and garlic) only on Plots 1 and 2. The crops shared in both land-units were parameterized with the same input-output inventory, reflecting the relatively homogeneous management and performance of candidate crops observed across all five plots (Table S1). All crop input-output data were collected from farmers' interview and local cropping system survey, representing the average level in the region. At this stage, we did not consider the inter-annual uncertainties of crop performance.

We explored future three-year farm crop sequence plans to enhance sustainability of farm production. A three-year planning window is generally acceptable for farmers in the North China Plain, where long-term crop sequence planning is uncommon (Xiao et al. 2022). In this context, crop sequences longer than three years are technologically complex and less practical for farmers. Importantly, even within a short planning horizon, whole-farm crop sequencing offers significantly greater opportunities for crop diversification compared to the prevailing wheat-maize annual double cropping system.

The planning period in this study was set to one year as managers often assess farm performance on an annual basis. The starting date of planning was set at October 1st, immediately following the harvesting of summer maize. Moreover, we considered the historical crop sequences on each plot over the past two years.

We chose wheat-maize double cropping in each year as the historical and future baseline crop sequence across the five plots. Following farmers' preference, Plot 3 was dedicated solely to the wheat-maize system throughout the planning window. As a result, while Plot 3 remained part of the sustainability assessment (sharing land-unit data with Plots 4 and 5), it was excluded from the crop sequence generation process.

3.2.2 Sustainability objectives

We set five sustainability indicators as objectives to improve at the whole farm level by designing diversified farm cropping plans, containing one from the economic domain (gross margin) and four from the environmental domain (groundwater depletion, pesticide input, eutrophication potential, and GHG emission).

Maximizing farm gross margin ($I_{GM,farm}$, USD/ha/year) over the planning window is the key interest of the farmers. The model evaluated the farm gross margin using Eq. 3, based on the gross margin ($Ind_{GM,c}$, USD/ha) of each component crop in the farm, which was calculated as:

$$I_{GM,p,t,c} = Y_{p,t,c} \times PR_c - CO_{p,t,c} \tag{11}$$

Where $Y_{p,t,c}$ (Mg/ha) and PR_c (USD/Mg) is the yield and sales price of the harvested commodity for crop c. $CO_{p,t,c}$ (USD/ha) is the variable costs for agricultural inputs (i.e., seeds/seedlings, irrigation, fertilizers, animal manures, pesticides, plastic films, and plant growth regulators.), hired labor, and machinery required for crop c (Table S1).

Minimizing groundwater depletion ($I_{GWD,farm}$, mm/year) plays a critical role in alleviating the groundwater crisis in the North China Plain and is of key interest to the county government. The model estimated this indicator using the following equation:

$$I_{GWD,farm} = I_{GC,farm} - D_{farm} \tag{12}$$

Where $I_{GC,farm}$ (mm/year) means farm-level groundwater consumption for crop production, which is derived from the groundwater irrigation amount used for each component crop ($I_{GC,p,t,c}$, mm) in a farm plan (Table S1):

$$I_{GC,farm} = \frac{\sum_p \sum_t \sum_c (I_{GC,p,t,c} \times A_p)}{\sum_p A_p \times PW} \tag{13}$$

Table 1 The list of candidate crops and their agronomic constraints and suitable plots for the case farm in this study. Max. Fre. means the maximum planting frequency of a certain crop in a crop sequence; The planting frequency of a certain crop is defined as the ratio of the number of times the crop is planted in a crop sequence and the length of the sequence. Min. Rep. (in years) means the minimum time interval between the repeated cultivation of a certain crop in a crop sequence.

No.	Crop name (abbreviation)	Latin name	Botanical family	Planting date	Harvest date	Max.Fre.	Min.Rep.	Suitable plots
1	Wheat	*Triticum aestivum*	Gramineae	Oct.15	Jun.1	1	0	1, 2, 3, 4, 5
2	Summer maize (SuMaize)	*Zea mays*	Gramineae	Jun.10	Oct.1	1	0	1, 2, 3, 4, 5
3	Spring maize (SpMaize)	*Zea mays*	Gramineae	May.20	Oct.1	1	0	1, 2, 4, 5
4	Peanut	*Arachis hypogaea*	Fabaceae	Apr.15	Aug.25	0.5	1.5	4, 5
5	Sunflower	*Helianthus annuus*	Asteraceae	Feb.25	Jul.1	0.5	1.5	1, 2, 4, 5
6	Potato	*Solanum tuberosum*	Solanaceae	Mar.5	Jun.20	0.5	1.5	1, 2, 4, 5
7	Sweet potato (SwPotato)	*Ipomoea batatas*	Convolvulaceae	Apr.12	Sep.30	0.5	1.5	1, 2, 4, 5
8	Garlic	*Allium sativum*	Amaryllidaceae	Oct.5	Jun.3	0.5	1.5	4, 5

While D_{farm} (mm/year) means the farm-level deep infiltration, estimated as:

$$D_{farm} = 0.02 \times (I_{GC,farm} + AP) \quad (14)$$

Where AP (mm/year) is the annual precipitation amount and 0.02 is the unitless recharge coefficient for the study area.

Minimizing farm pesticide input ($I_{Pesti,farm}$), expressed in kg active ingredient (AI)/ha/year, is strongly emphasized in the national and regional agricultural policies on the reduction in agrochemicals. The pesticide input at the farm level was also aggregated from the crop-level input ($I_{Pesti,p,t,c}$, kg AI/ha/year) using Eq. 3. $I_{Pesti,p,t,c}$ was computed as follows:

$$I_{Pesti,p,t,c} = I_{Insect,p,t,c} + I_{Fungi,p,t,c} + I_{Herb,p,t,c} \quad (15)$$

Where $I_{Insect,p,t,c}$, $I_{Insect,p,t,c}$ and $I_{Insect,p,t,c}$ (kg AI/ha/year) denotes insecticide, fungicide, and herbicide input for crop c, respectively (Table S1).

Minimizing life-cycle aquatic eutrophication potential ($I_{EUP,farm}$) and GHG emissions ($I_{GHG,farm}$) indicate combating nutrient pollution and global warming. These two indicators accounted for environmental impacts from both pre-farm and on-farm stages. The pre-farm stage includes corresponding impacts associated with the manufacturing and transportation of agricultural inputs such as fertilizers, pesticides, electricity for irrigation, and fuel for machinery use. The on-farm stage encompasses environmental impacts induced by field operations. Likewise, their farm-level assessment was done using Eq. 3 with corresponding crop-level performance ($Ind_{EUP,p,t,c}$, kg PO_4 eq./ha/yr and $Ind_{GHG,p,t,c}$, Mg CO_2 eq./ha/yr):

$$I_{EUP,p,t,c} = I_{EUP-PRE,p,t,c} + I_{EUP-ON,p,t,c} \quad (16)$$

$$I_{GHG,p,t,c} = I_{GHG-PRE,p,t,c} + I_{GHG-ON,p,t,c} \quad (17)$$

Where $I_{EUP\text{-}PRE,p,t,c}$ and $I_{EUP\text{-}ON,p,t,c}$ (kg PO_4 eq./ha/yr) represent pre-farm and on-farm eutrophication potential for crop c, while $I_{GHG\text{-}PRE,p,t,c}$ and $I_{GHG\text{-}ON,p,t,c}$ (Mg CO_2 eq./ha/yr) represent pre-farm and on-farm eutrophication potential for crop c. The calculation for these pre- and on-farm environmental performances is further detailed in the Supplementary Material S1.

3.2.3 Constraints

From the perspective of longitudinal constraints (temporal-scale constraints), we formulated multiple crop sequence rules with local agronomists to filter out agronomically infeasible sequences but did not set any plot-level constraints that would define acceptable ranges of indicator values as the farmer did not have specific demands in the regard. Crop sequence rules we used included the maximum frequency of a crop or crop group in a sequence, the minimum interval between repeated cultivation of a crop or crop group in a sequence (Table 1), and disallowed crop successions (Table S2). To align with local widespread practices, we permitted cereals to be planted every year because they were quite tolerant to continuous cropping, while other crops sensitive to continuous cropping, such as peanut, potato and garlic, had to be cultivated every other year to avoid substantial yield penalties. Crop successions were disallowed when crops were from the same botanical family or there was a long fallowing period between them.

Farm managers proposed several transversal constraints (spatial-scale constraints) on indicator performance and crop harvested area to fit generated farm plans with their demands. At first, we defined two indicators that were used as period- and farm-level constraints in the model: labor use and dietary energy yield. The labor use per crop ($I_{Labor,p,t,c}$, h/ha) was quantified as (Table S1):

$$I_{Labor,p,t,c} = \sum_f I_{Labor,p,t,c,f} \quad (18)$$

Where f denotes the manual farming operations including transplanting, fertilization, irrigation, pesticide application, pruning, and harvesting. The crop-level dietary energy yield ($I_{DE,p,t,c}$, persons/ha) was estimated as

$$I_{DE,p,t,c} = \frac{Y_{p,t,c} \times F_c}{RAI} \quad (19)$$

Where F_c is the fraction of dietary energy in the product of crop c (Table S3). RAI is the recommended annual intake of dietary energy of a reference person (Yang et al.. 2018). Their period- and farm-level performance was computed using Eqs. 2 and 3.

Then multiple period-level constraints of indicator performance were put forward: 1) The gross margin in each period must be higher than the baseline; 2) The groundwater consumption, pesticide input, eutrophication potential, and GHG emission must be lower than the baseline in each period; 3) The total labor use (hired plus family labor) had to not exceed 520 h in each period. With these constraints, generated farm plans could be used to find ways to improve five sustainability objectives at once. Furthermore, three constraints on crop harvested area should be met: 1) Wheat area must be greater than 8 ha per period to fulfill the supply contract that managers signed with a flour mill; 2) Maize area should not exceed 12 ha per period to prevent the dominance of maize production; and 3) The harvested area for sunflower, potato, and sweet potato must be limited to less than 8 ha due to limited market access. Regarding the whole-farm level, we required the dietary energy yield to be at least half of the baseline plan to rule out farm plans

that were unproductive in food energy. These constraints collectively guide the generation of farm cropping plans that are both economically viable and environmentally friendly, meeting the specific needs and constraints of the case farm.

4 Results

4.1 Farm baseline plan versus Pareto-optimal plans

The baseline crop sequence plan for the case farm, i.e., continuous wheat-maize double cropping over future three years across all plots, resulted in a farm gross margin of 2303 USD/ha/yr, groundwater depletion of 284 mm/yr, pesticide input of 6.67 kg AI/ha/yr, eutrophication potential of 28.9 kg PO_4 eq./ha/yr, and GHG emission of 10.0 Mg CO_2 eq./ha/yr (Fig. 5).

FarmSTEPS generated 213,679 feasible farm-plans, with 128 of them being Pareto-optimal in terms of five sustainability objectives. Compared with the baseline farm plan, these Pareto-optimal farm-plans achieved 3–50% higher farm gross margin while reaching 32–56% lower groundwater depletion, 31–47% lower pesticide input, 10–33% lower eutrophication potential and 22–40% lower GHG emission. Thus, the diversification of crop sequences on the case farm resulted in improvements of all five indicators, though trade-offs existed between economic and environmental objectives (Fig. 5).

The improved sustainability performance of optimal farm-plans was mainly due to the inclusion of non-cereal crops, particularly sweet potato and sunflower (Table 2). On the largest plot in the farm, Plot 4, half of the optimal farm-plans suggested a sweet potato-based crop-sequence (SwPotato-Wheat-SuMaize-SwPotato), and the other half combined sweet potato and sunflower with wheat-maize double cropping. On the intermediate-size plots, Plot 2 and 4, most generated crop-sequences were still combinations of sweet potato, sunflower, and wheat-maize double cropping in the crop sequence, accounting for 37% and 64% of optimal plans for two plots respectively. However, the model generated garlic-maize double cropping for Plot 2, a plot in which labor-intensive crops were allowed to be planted, in 36% of optimal plans. The generated crop-sequences on Plot 1, the smallest plot, offered more space for crop diversification across optimal plans, with only 5% involving wheat-maize double cropping.

4.2 Example farm plans

We selected and presented three individual plans as examples from the Pareto-optimal set of 128 farm-plans (Fig. 6). These three example farm-plans showed contrasting sustainability profiles, including one profit-prioritized farm plan, one environment-prioritized plan, and one plan that achieved a balance between profitability and environmental soundness (Fig. 5; Fig. 6).

The profit-prioritized farm-plan created the highest farm gross margin, 50% higher than the baseline, while also incurring the highest environmental costs across optimal plans, but still below the baseline (Fig. 5; Fig. 6). Such performance resulted from the crop-sequence on the largest plot (Plot 4) being dominated by sweet potato, the most profitable crop in this plot, as well as crop-sequences that frequently cultivated profitable crops, especially garlic, on the intermediate- and smallest-size plots (Plot 1–2, Fig. 6). In contrast, the environment-prioritized plan was one of the least environmentally impactful plans across the Pareto set but also the third least profitable one. Compared with the baseline, this plan reduced environmental costs by 32–56% while increasing gross margin by only 9%. The success of its environmental performance was in part due to the cultivation of sunflower, the most resource-saving crop, and the markedly decreased land-use intensity realized by replacing wheat-maize double cropping with low-input single crops in a year, such as sweet potato, peanut, and spring maize (Fig. 6).

The balanced farm-plan resulted in an intermediate level of sustainability performance between the profit- and environment-prioritized farm-plan. Similar to the profit-prioritized farm-plan, its crop sequence in Plot 4 still included sweet potato, while the crop-sequences on the intermediate- and smallest-size plot presented a mix of profitable crops, e.g., garlic and potato, and less resource-intensive crops, e.g., sunflower and sweet potato.

5 Discussion

In this study, we presented a novel rule-based model, FarmSTEPS, for exploring sustainable and diversified farm crop sequence plans aligned with farm-specific situations and constraints. This modelling framework integrates farm characterization modules, an algorithm for exhaustive whole-farm crop sequence planning, and a multi-attribute decision-aid tool that can recommend a set of desirable farm cropping plans in a systematic and spatial-temporally explicit manner. We applied FarmSTEPS in a case farm and successfully obtained 128 optimal three-year crop sequence plans, which markedly outperformed the baseline yearly repeating wheat-maize double cropping plan in gross margin, groundwater depletion, pesticide input, eutrophication potential, and GHG emission concurrently, albeit with trade-offs occurring between economic and environmental objectives. The plot-specific crop sequences of these optimal plans highlighted the important role of a few of candidate crops, e.g., sweet potato

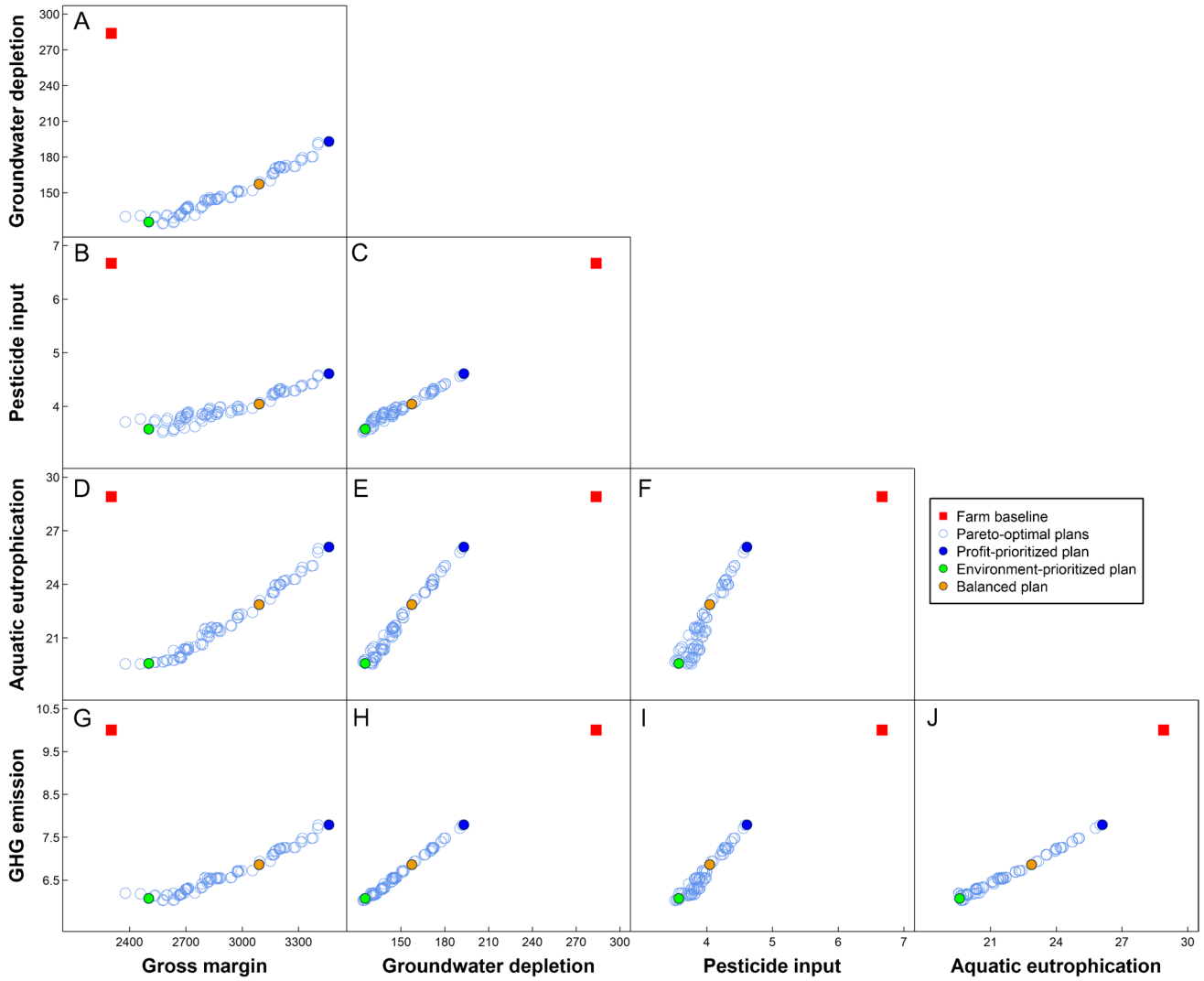


Fig. 5 Biplot matrix of sustainability objectives across 128 Pareto-optimal farm cropping plans (in blue circles) and baseline farm plans (the single red square). Sustainability objectives comprised gross margin (USD/ha/yr), groundwater depletion (mm/yr), pesticide input (kg AI/ha/yr), aquatic eutrophication (kg PO_4 eq./ha/yr), and GHG emission (Mg CO_2 eq./ha/yr). Three examples of optimal plans were highlighted in blue, orange, and green, representing profit-prioritized, balanced, and environment-prioritized farm plan, respectively.

and sunflower, in advancing overall sustainability. Thus, the model outcomes do not only present the potential and trade-offs of sustainability objectives but also identify promising crops and crop sequences for the studied farm, which could support the stakeholders' discussion to facilitate informed decision-making on future farm production choices. While demonstrated on a single case farm, the FarmSTEPS model serves as a flexible framework that can be adapted to diverse crop farming contexts by adjusting plot configurations, candidate crop data, farm-specific constraints and objectives, thus supporting the navigation of sustainable cropping solutions tailored to different farm circumstances and aspirations.

5.1 Model-based design of diversified whole-farm cropping plans towards multi-attribute sustainability

The design of sustainable and diversified crop sequence plans for farms is a complex process, entailing a combination of many elements: plot configuration and heterogeneity, various candidate crops, various sequence- and farm-level constraints, numerous decision-making periods, and multiple sustainability goals. We developed the FarmSTEPS model to assist farm stakeholders to systematically generate

Table 2 Top five crop sequences planned for each plot across 128 Pareto-optimal farm plans. Plot 3 was set to be dedicated to wheat-maize double cropping over the planning window. SuM, SpM, and SwPot mean summer maize, spring maize, and sweet potato.

Plot number	Crop sequence in the planning window			Frequency
	Period 1	Period 2	Period 3	
Plot 1	Peanut	SwPotato	Sunflower	13
	Garlic-SuMaize	SwPotato	Garlic-SuMaize	12
	SpMaize	SwPotato	Sunflower	11
	Garlic-SuMaize	SwPotato	Sunflower	10
	Garlic-SuMaize	SwPotato	Potato	9
Plot 2	Wheat-SuMaize	SwPotato	Sunflower	17
	SwPotato	Sunflower	Wheat-SuMaize	15
	Sunflower	Wheat-SuMaize	Wheat-SuMaize	15
	Garlic-SuMaize	Sunflower	Wheat-SuMaize	11
	Garlic-SuMaize	Sunflower	Garlic-SuMaize	10
Plot 3	Wheat-SuMaize	Wheat-SuMaize	Wheat-SuMaize	128
Plot 4	SwPotato	Wheat-SuMaize	SwPotato	60
	Wheat-SuMaize	SwPotato	Sunflower	25
	Sunflower	Wheat-SuMaize	SwPotato	21
	SwPotato	Sunflower	Wheat-SuMaize	21
	Wheat-SuMaize	Sunflower	Wheat-SuMaize	1
Plot 5	Wheat-SuMaize	SwPotato	Sunflower	46
	SwPotato	Sunflower	Wheat-SuMaize	19
	Sunflower	Wheat-SuMaize	SwPotato	17
	Wheat-SuMaize	SwPotato	Wheat-SuMaize	15
	Wheat-SuMaize	Sunflower	Wheat-SuMaize	9

promising farm plans with a given set of abovementioned elements, without missing any possible solutions.

Many of existing whole-farm planning models, such as FarmIMAGES (Dogliotti et al. 2005) and FarmDESIGN (Groot et al. 2012), also aim to address these challenges, albeit through different methodological approaches. FarmIMAGES identifies more sustainable cropping plans by first generating multi-year crop rotations at the plot scale, followed by a multi-objective mixed-integer programming algorithm to allocate these rotations across fixed plots (Dogliotti et al. 2014). This structure enables the efficient selection of a single optimal plan that enhances mean annual economic and environmental performance at the whole-farm level. While this approach is computationally efficient and robust, its single-solution outcome may overlook alternative plans that align with diverse stakeholder priorities. In contrast, FarmDESIGN uses a heuristic algorithm to allocate crop rotations based on proportional farmland area, resulting in a set of near-optimal solutions (Ditzler et al. 2019). Its flexible approach, allowing crop allocation without fixed plot boundaries, can reveal new opportunities for improving farm-scale sustainability outcomes. However, due to the nature of heuristic optimization, the solutions may be non-robust in some cases and may converge to local optima, especially when the number of crop rotation options is large. FarmSTEPS, by comparison, takes a comprehensive and exhaustive approach: it first enumerates all agronomically feasible crop sequences at the plot level, and then systematically constructs all possible whole-farm plans across the planning horizon by combining these sequences over time and space, while incorporating spatial and temporal constraints. This integration of exhaustive enumeration with constraint-guided filtering allows the model to spatial-temporally explicitly generate and preserve all promising plans while aligning them with real-world sustainability goals and stakeholder preferences. Through Pareto-based evaluation, FarmSTEPS effectively narrows the solution space and provides a set of desired farm plans against multiple sustainability objectives. Despite the heavier computational burden and limited flexibility in modifying plot structures within the model, these challenges can be fully or partially addressed by incorporating appropriate constraints and importing revised farm plot maps that reflect the desired configurations.

Compared with existing models, one of the primary advancements of FarmSTEPS is its ability to assess farm-scale sustainability performance and enforce production constraints at finer temporal scales, such as on an annual basis within a multi-year planning horizon. For example, many models do not account for year-by-year gross margins or allow for the specification of minimum gross margin thresholds per year. Cropping plans with high mean annual returns but extremely low margins in individual years may

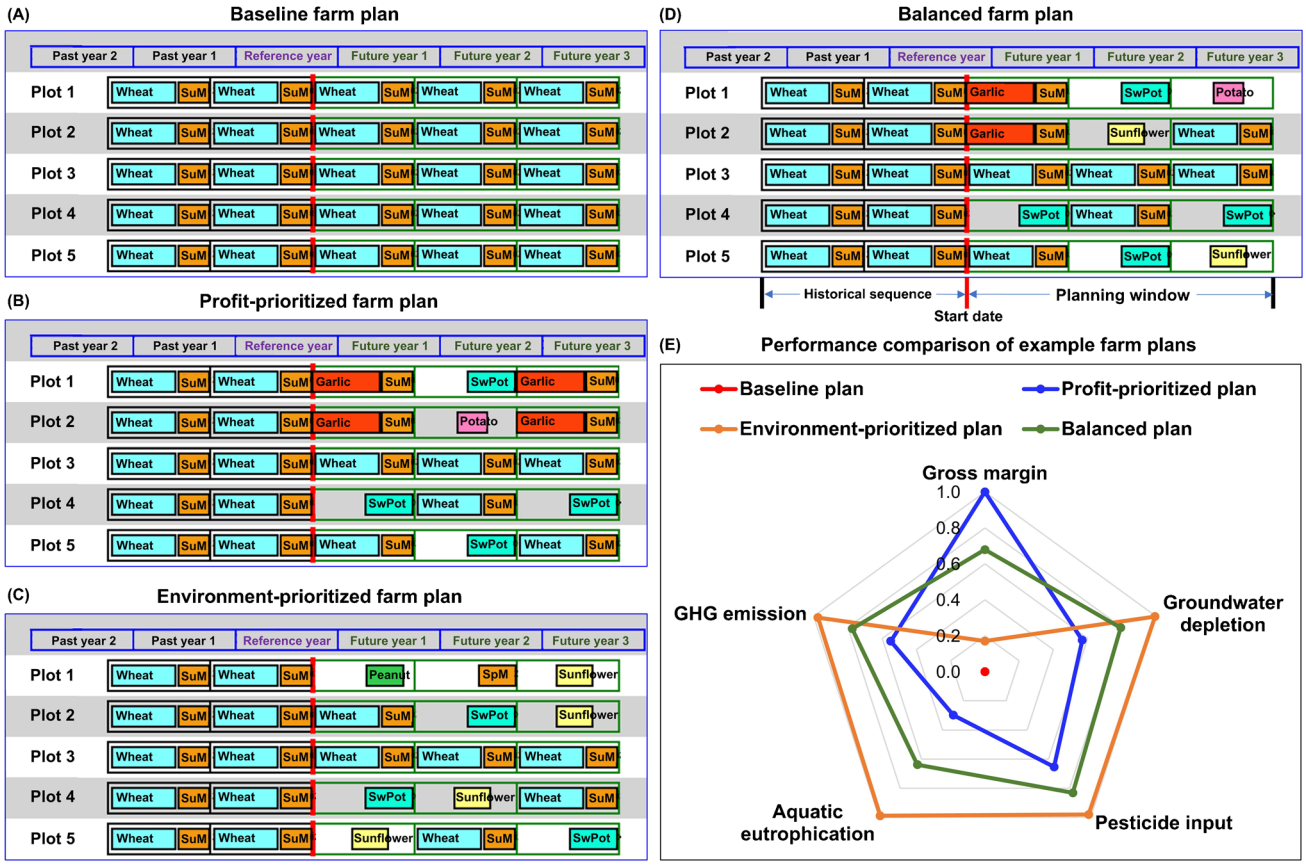


Fig. 6 Spatial-temporal crop allocation (**A**-**D**) and sustainability objective performance (**E**) of baseline and three examples of optimal farm-plans with contrasting sustainability profiles. The top row in panel **A**-**D** indicates the calendar year, and the reference year means the calendar year of the start date. The length of the box with crop names represents the growing season of the crops, and the breaks (white sections) between two adjacent crops represent fallow periods. SuM, SpM, and SwPot mean summer maize, spring maize, and sweet potato. In the radar plot, indicator values were rescaled to 0–1 using Max-Min approach where 0 and 1 denote the worst and best performance observed over all Pareto-optimal and original farm plans. The figure illustrates trade-offs between economic performance and environmental performance among the optimized crop sequence plans.

still be impractical for farmers who must manage cash flow and financial risk. In this regard, FarmSTEPS introduces a novel combination of spatial and temporal design and evaluation, enabling the creation of diversified, sustainable cropping plans that are better aligned with the operational realities and long-term goals of farmers.

Another innovative feature of our model is its ability to generate flexible and non-cyclical crop sequences, rather than being limited to fixed crop rotations, for farm plots over the planning window (Castellazzi et al. 2008). Flexible crop sequences better represent the actual farmers' decision-making tactics and offer greater opportunity to improve farm performance. For instance, in our case farm, the model created a flexible three-year sequence of (SwPotato-Wheat-SuMaize-SwPotato) for the largest plot (Plot 4) in almost half of optimal farm plans, which greatly contributed to better economic and environmental outcomes. However, if the model is restricted to generating only cyclical three-year rotations, this sequence would be disallowed because sweet potato is prohibited to be planted after itself according to predefined agronomic rules (Table S2), thereby limiting the sustainability benefits of new plans. Although flexible sequences might increase the management complexity for farmers, this feature can help them explore the full range of future possibilities and potential improvements.

In our case study, the model produced a range of optimal farm cropping plans that reached better economic and environmental sustainability benefits than the baseline wheat-maize plan, and did so for all indicators. The resulting crop sequences showed that wheat-maize double cropping was mainly substituted by sweet potato and sunflower in optimal plans, particularly on the larger-size plots. This substitution can be explained by the fact that these two crops were resource-efficient, well mechanized, and moderately productive in dietary energy with comparable or higher profitability

than the currently dominant wheat-maize system (Table S1), thus offering ample chances to improve sustainability objectives while adhering to constraints regarding labor availability and dietary energy provisioning. Without the inclusion of such crops in the candidate crop list, the model would probably not be able to create farm plans outperforming the baseline across all objectives. In such cases, relaxing constraints on indicator values would be necessary to allow the model to produce feasible farm plans, but at the expense of either economic or environmental benefits associated with the adoption of the new farm plans in comparison to the reference plan. Consequently, the model outcomes underscored the vital role of crops possessing the aforementioned desired attributes in future crop farming systems, particularly for intensified and specialized food production areas where labor resources are scarce.

In the future, multi-actor participatory approaches are needed to bridge the gap between model outcomes and final decision-making for real-world application (Leclère et al. 2023). Engaging relevant stakeholders, such as farmers, supply chain actors, and policymakers is crucial to identifying farm cropping plans that achieve compromises across competing objectives and are socially and institutionally acceptable (Liang et al. 2026). For example, group workshops can provide a platform for stakeholders to collectively evaluate the feasibility, benefits, and potential limitations of selected plans (Quinio et al. 2022). Such workshops could help identify and improve compromise solutions that cater for the demands of multiple actors and promote the translation of model outputs into actionable and context-sensitive decisions for implementation.

5.2 Opportunities for widespread application of the FarmSTEPS model

The FarmSTEPS model holds strong potential for broad application in exploring sustainable and diversified farm cropping plans across different world regions. Its flexibility in dealing with different farm settings, crop portfolios, production constraints, and sustainability objectives makes it well-suited to a variety of agricultural contexts. The model can be adapted for use on farms that vary by crop type (from arable farms and horticultural farms, to grassland farms), production scale (from smallholder farms to large-scale farms), and management modes (from organic to industrial). However, successful application requires necessary adaptations of model inputs to reflect specific farm conditions.

These adaptations fall into five main aspects:

1) Farm plot configurations: Users can extract plot map from satellite or aerial imagery and convert them into ESRI files that records geographical attributes of each plot such as location, area, and slope.
2) Sustainability objectives: Farm stakeholders may opt for varying sets of sustainability indicators to serve as objectives in cropping plan design. For instance, subsistence smallholders may prioritize the production of dietary nutrients ranging from calorific energy and protein to minerals and vitamins (Timler et al. 2020), and farms in mountainous areas focus on lowering soil erosion (Smith et al. 2017). FarmSTEPS allows users to flexibly define or select any indicators computable from by crop and plot parameters within defined land-units.
3) Land-units and candidate crops: Users can define the land-units for each of farm plots, where soil properties, resource availability, and candidate crops for each plot are specified according to the actual situation. The selection of candidate crops is guided by factors such as compatibility with pedo-climatic environments, market orientation, and farmers' preferences (Adhikari et al. 2023). For instance, horticultural farms may include only vegetables, fruits and flowers as candidates. Crucially, users must collect crop data required for the assessment of farm-specific sustainability objectives. Such input-output data of candidate crops should reflect their performance under the farming context of the focus farm or plot. For example, for a particular crop, higher yield and water usage on irrigated plots versus rainfed plots, and lower yield and less synthetic agrochemical inputs in organic farms versus conventional farms. The input-output inventory of crops can be compiled based on a variety of sources, including local farm surveys, expert knowledge, statistical documents, and scientific literature (Liang et al. 2023).
4) Production constraints: The longitudinal and transversal constraints can be tailored to farm's unique barriers and demands. Longitudinal constraints, i.e., crop sequence rules, may be formulated based on local agronomists and local field observations. Transversal constraints can be defined not only in terms of total requirement and harvested area of crops per unit planning period, but also input or indicator thresholds. For example, annual irrigation water quotas may apply in farms in water-scarce regions (Zhang and Oki 2023), while fertilizer quotas could be imposed for farms in nutrient-surplus regions (Expósito and Velasco 2020).
5) Planning timeline: Timeline parameters can be determined with farmers to reflect their actual decision-making behaviors and account for the farm reality. For instance, the unit planning period may exceed one year to account for perennial crops on grassland farms. Organic farms, which often adopt long-term crop sequences, may benefit from planning windows that span a decade or more.

With above adaptations, our model can create a set of tailored cropping plans for individual farms, providing stakeholders with valuable insights into the potential benefits and trade-offs across multiple sustainability objectives. Widespread adoption of the model among farms in a region could also help policymakers understand the contribution of crop diversification to combating regional sustainability challenges and the intractable trade-offs across objectives most farms perhaps encounter. This, in turn, can assist policymakers in identifying macro-level solutions, e.g., incentivizing machinery specific to certain crops, to further foster agricultural sustainability.

Lastly, while the model is flexible and versatile, FarmSTEPS is specifically designed for whole-farm crop sequence planning and does not currently account for other dimensions of farm management, such as livestock systems, by-product allocation, or landscape-level modifications.

5.3 Limitations

Several limitations of the model may need to be addressed in future work. First, we did not consider the uncertainty in crop yields, market prices, and weather over the planning window (Komarek et al. 2020), as the current version of FarmSTEPS is deterministic. In this case study, we used crop input-output data collected in a year with a normal weather pattern and average price data over the past five years to mitigate the effects of uncertainty. Moving forward, we may think of introducing stochastic parameters, allowing users to input a reasonable distribution of parameters rather than fixed values (Ballot et al. 2018). This will enable us to examine the effects of future uncertainty on the performance of whole-farm plans, though generating outcomes would further aggravate the computational burden of obtaining the solutions and their performance.

Furthermore, FarmSTEPS did not account for crop interactions at temporal and spatial scales. For example, the yield of a crop may be influenced by preceding crops in a crop sequence (Reckling et al. 2016), and farmers should avoid planting some crops in neighboring plots to prevent spread of pests and diseases and associated yield loss (Juventia et al. 2022). Incorporating pre-crop effects and crop topological constraints into the next version of FarmSTEPS will enhance the accuracy of farm assessments.

FarmSTEPS serves as a valuable tool for exploring trade-offs across multiple sustainability objectives in the redesign of farm cropping plans. However, the current modeling framework does not yet incorporate quantitative trade-off analysis methods, such as sensitivity analysis and shadow price analysis (Kanter et al. 2018). Sensitivity analysis can help identify how small changes in model inputs (e.g., resource availability, crop yields, or market prices) influence outcomes across different objectives, thereby highlighting which parameters most affect the trade-off dynamics. Shadow price analysis estimates the marginal cost or benefit of improving one objective at the expense of another, offering stakeholders a clearer picture of the 'price' of trade-offs (Breure et al. 2024). Integrating such methods into future applications of FarmSTEPS could provide a deeper understanding of the drivers and intensities of trade-offs, thereby enhancing transparency and supporting more informed, evidence-based decision-making.

Lastly, FarmSTEPS remains challenging for farmers to use independently. The first barrier is its application involves extensive data collection and specialized tasks such as farm mapping and multi-criteria assessment. Therefore, at this stage, agricultural advisors are crucial in working with farmers to use the model and explore more sustainable farming options. Moreover, running the model can require high-performance hardware in some cases. In our case study, a farm with five plots, eight candidate crops, and three planning periods required approximately 15 minutes for FarmSTEPS to generate all feasible farm plans. As the number of land parcels, alternative crops, and planning horizons increases, high-performance computing platforms may be needed to complete the crop planning process efficiently. In such situations, IT specialists are essential to support farmers in model implementation, computation management, and result interpretation.

6 Conclusion

In this study, we presented a novel farm planning model, FarmSTEPS, built to systematically explore sustainable and diversified crop sequence plans for crop farms in a spatial-temporally explicit manner. The model can capture spatial configuration, plot heterogeneity, production constraints, and development objectives specific to each farm, thus ensuring that generated plans are tailored to each farm's unique conditions. We illustrated it using a wheat-maize farm in China, where the model created a set of optimal plans superior to the baseline plan in both economic and environmental performance, while also uncovering a trade-off between these two sustainability domains. FarmSTEPS can assist in identifying the critical crops and crop sequences specific to each plot for sustainable farm production. Users can customize the model for farms of interest according on-farm realities, which offers opportunities for widespread application of our model across different farming regions. Overall, FarmSTEPS serves as an efficient tool for designing sustainable whole-farm cropping plans and informing stakeholders about synergies and trade-offs among sustainability objectives.

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Data availability All data used in this study are included in this published article and supplementary materials.

Code availability Software and data can be downloaded from: https://git.wur.nl/jeroen.groot/farmsteps.package.

Declarations

Ethics approval The study was performed in accordance with the ethical standards as laid down in the 1964 Declaration of Helsinki.

Consent to participate Informed consent was obtained from all individual participants in this study.

Consent for publication Informed consent was obtained from all individual participants in this study to publish the data as presented in this manuscript.

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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